

# Ihtisham Ali

## Congituve RE for AI systems

 Thesis Evaluation

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## Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology

### Synopsis for the Degree of Master of Science (MS)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                     |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--|
| Name of Student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Sabahat Siddique                                                                                                    |  |
| Department                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | School of Computing Sciences                                                                                        |  |
| Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MS Artificial Intelligence                                                                                          |  |
| Registration No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M24F0023AI003                                                                                                       |  |
| Name of Research Supervisor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Dr. Musadaq Mansoor                                                                                                 |  |
| Co-Supervisor (if any)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Dr. Maqbool Khan                                                                                                    |  |
| Other members of the GEC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1. Dr. M. Shuaib Qureshi<br>2. Dr. Asim Farooq                                                                      |  |
| Title of Research Proposal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CREAIS: Cognitive Requirement Engineering for the Next Generation of Ethical Artificial Intelligence Systems</b> |  |
| <b>Summary of the Research Proposal</b><br>Artificial intelligence (AI) technologies are increasingly used in sensitive areas of criminal justice, healthcare, banking, and self-driving cars. Despite the numerous benefits of such technologies, it also raises some grave ethical concerns including discrimination, discrimination, lack of transparency, threat of privacy, and lack of responsibility. Although functional and quality requirements are the primary concerns of the traditional requirements engineering (RE) |                                                                                                                     |  |

methodologies, ethical concerns are hardly ever consistently integrated in system development. It is due to that fact that such moral principles as justice, explicability, and reliability often remain abstract and hard to implement. To overcome this issue, it is required to develop a specifically designed AI-based RE framework that would have both incorporated ethical considerations at an early stage of system design and ensured that ethical standards would be verified through the development lifecycle.

The proposed research offers a larger RE framework to AI systems through the addition of ethical considerations to elicitation, analysis, specification, validation, and traceability stages of the process. It will be possible to convert the ethical principles of fairness, responsibility, privacy, and explicability into how quantifiable parameters and be able to validate them using benchmark datasets (including UCI Adult Income, COMPAS, and German Credit) and interpretable models (including logistic regression and decision trees). A web-based application will be developed to record the ethical requirements, compare them using fairness and explainability metrics, and maintain a trace between the datasets, models and the evaluation outcomes. The expected input of this project is a helpful framework and validation platform that enhances the openness, responsibility, and moral soundness of AI development.

## **Introduction**

The use of artificial intelligence (AI) technologies in such a field as health care, banking, criminal justice, and self-driving cars is becoming increasingly frequent. Such systems are actively used to support or replace the human decision-making in the field of employment where the social and ethical consequences are pressing. Efficiency and cost-saving, scalability are some of their potential advantages, but the issues of bias, unfairness, invasion of privacy, lack of transparency, and accountability are very critical [1], [2]. Unregulated AI can enhance discrimination and lack of trust, which is manifested in the popular cases of bias in recidivism prediction and credit score systems [3]. To mitigate such issues, the ethical issues must be addressed as a regular aspect during the construction of AI systems rather than being viewed as an accidental factor.

Decision of what a system should do and with what constraints is one of the significant tasks of requirement engineering or RE. RE has conventionally paid more attention to its functional requirements i.e. capabilities of the system, i.e. quality criteria i.e. performance, usability, and security. Nonetheless, as soon as AI systems start influencing individuals and society, in general, RE will be forced to be modified in a manner that would accommodate the ethical standards like



responsibility, privacy, explainability, and fairness [1], [4]. The fact that the ideas of ethical practices need to be transformed into more quantifiable, visible and traceable practices is a major issue because such ideas have the tendency to be abstract and circumstantial. The gap between the ethical ideals on the high level and its implementation in the creation of AI is explained by the fact that the modern processes and technologies of RE do not tend to give systematization in addressing these issues [2], [5].

It has a lot of research studies, which have identified this problem as well as given some partial solutions. It has been observed that despite all the engagements, not all the ethical issues such as openness and fairness are always incorporated into the processes of RE [1]. According to some researchers, ethical requirements of AI systems ought to be considered similarly to the legal requirements that are based on the codes and principles [2]. In a bid to formalise and certify ethical requirements, assurance-case-based models have been proposed [7] and ontology-based modelling [6]. The explainability need has as well been examined using the user-centered ways [8], [9]. Cumulatively these studies demonstrate the significance of the ethics consideration in the process of RE, however, there is a lack of lifecycle traceability, tool support and quantifiable considerations. Our research fills these gaps by publishing AI-specific RE framework that will systematically consider the ethical considerations in the lifecycle of the development process. The framework will also incorporate the ideas of ethics as being part of the first-class requirements into the normal RE processes of elicitation, analysis, specification, validation, and traceability. With the help of structured templates and benchmark data like UCI Adult Income, COMPAS, and German Credit, even such abstract notions as fairness, privacy, accountability and explicability will be transformed into quantifiable requirements. The compliance with ethical standards will be checked with the help of interpretable models (Decision Trees and Logistic Regression) and explainability tools (SHAP and LIME). A web based validation tool will also be designed to record, test and trace requirements throughout the lifetime as a way of establishing accountability and constant monitoring. The study will seek to fill the gap between theoretical ethical principles and the real world of AI engineering through the approach and act as a practical framework and tool.

### **Problem Statement**

The current Requirement Engineering (RE) methods of Artificial Intelligence (AI) systems would not adequately characterize, operationalize, and validate ethical needs such as accountability, privacy, explainability, and fairness. The AI systems pose a threat to bias, obscurity, and lack of credibility since ethical standards are yet to be concrete and have no quantifiable parameters, tool support, and their lifecycle traceability.

### **Purpose of the Study**

The objective of this project is to design and introduce an AI-based Requirement Engineering (RE) framework that systematically incorporates such ethical requirements as responsibility, privacy, explainability, and fairness into all phases of AI system creation. To achieve this, quantifiable requirement templates will be established, operationalized with benchmark datasets and models, and a web based validation tool that ensures compliance to ethical standards, continuous testing and traceability will be generated.

### **Objectives**

1. To develop a Requirement Engineering (RE) framework that applies to AI which introduces the ethical limitations of privacy, responsibility, explainability, and fairness into the process of elicitation, analysis, specification, validation, and traceability in the development of AI systems.
2. To develop and implement a web-based validating tool based on the interpretable AI models and benchmark datasets that enables the recording, measure and continuous monitoring of ethical standards..

### **Research Questions:**

1. How do we incorporate the explainability, privacy, fairness, and accountability in the AI requirements?
2. How can an online tool be applied to monitor and observe ethical standards in the creation of AI?

### **Research Methodology**

Research methodology provides procedural steps and specified plan which will guide this research to achieve its objectives. Since the objective of the given research is to make Requirement Engineering (RE) accommodate ethical requirements to the AI systems, tool development, data-driven verification, and a framework ofiding development will be applied.

### **1. Review of Literature and Analysis of Gaps**

The first step is to review existing Requirement Engineering (RE) frameworks, the Responsible AI principles, and ethical requirement models in detail. This assessment will reveal limitations of current procedures, in particular, with respect to integration of ethical concepts such as accountability, privacy, explainability, and fairness. A gap analysis will be used to demonstrate where the existing solutions are inadequate to operationalize these principles into tangible needs to demonstrate the need of a new framework.

### **2. Development of Frameworks**

Depending on the findings of the literature research, an AI-specific RE framework will be developed. Ethical ideals will be defined by structured requirement templates that transform intangible ideas to quantifiable and testable components. These will be mapped to functional and quality requirements in order to ensure that ethics are systematically integrated to the elicitation, analysis, specification, validation, and traceability processes of the RE process. The framework will also align life cycle management processes with ethical standards to ensure that they remain sustainable during the evolution of the AI systems.

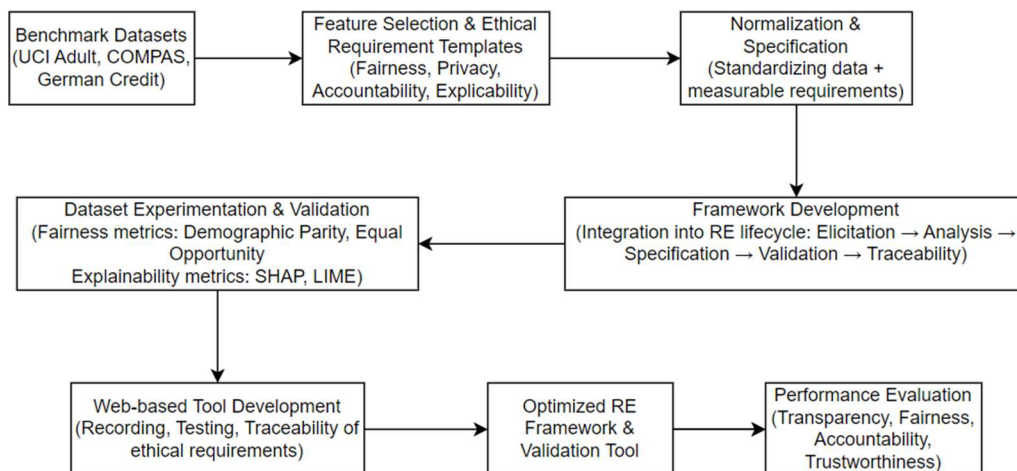
### **3. Validation and Experimentation with Datasets**

The operationalization of ethical standards will be based on benchmark datasets, including German Credit, Compas and UCI Adult Income. Such interpretable models like decision trees and logistic regression will be served as test cases and this will help in making the results clear and understandable. The compliance with the ethical standards will be achieved by using explainability

methods (e.g., SHAP, LIME) and fairness measures (e.g., demographic parity, equal opportunity). This action will show how the framework will turn the theoretical ideas into the practical deliverables of the real-life AI projects..

#### 4. Tool Creation and Application

A web-based validation tool will be developed to ease in the process of recording, testing and tracking of ethical criteria. Technology will also introduce some explainability and fairness features, which will enable AI models to be supervised constantly regarding the morals. It will also maintain traceability by relating requirements with information, models and evaluation results. The utility and effectiveness of the tool will be evaluated by the case studies that will also provide insights into how the tool will contribute to bridging the chasm between the theoretical models of RE and the development of the feasible AI systems.



Ac

#### Examples (as applicable)

##### Objective 1: Methodology

- The stages of AI system development elicitation, analysis, specification, validation, and traceability are complemented with the stage of ethical requirements in order to come up with an Requirement Engineering (RE) model, which will also be AI-oriented.

- To create loopholes, look at the literature of the existing RE frameworks, Responsible AI guidelines and models of ethical requirements.
- Definitely set ethical guidelines (explainability, privacy, responsibility, and fairness) with the help of systematic templates.
- Transform the moral values into measurable values based on interpretable models( Logistic Regression, Decision Trees ) and benchmark data( UCI Adult Income, COMPAS, German Credit ).
- These are requirements, which have to be considered during the formulation of the extended RE model, in his lifecycle phases.

### **Objective 2: Methodology**

- To design and deploy an online validation device and rely on interpretable AI models and benchmark data that would allow recording, measuring, and continuously track ethical standards.
- Design a web application that is traceable, has validation and requirement capturing modules.
- The tool contains fairness and explainability, such as equal opportunity, demographic parity, SHAP, and LIME.
- Benchmark Test the tool using benchmark datasets and compare model results.
- Show how the ethical principles may be tracked and verified in the course of the AI development in order to justify the technology.

### **Literature Review**

The bias in algorithms is yet another common problem discussed in the article published in [1] with severe ethical concerns of using AI to the healthcare, banking, and transportation industries. It indicates that it generates bias in the training data and design decisions and leads to discriminating mechanisms. The article does not present practical mitigation models, at the cost of the broad coverage of the identification of the problem. The future work is oriented to the establishment of the standardized protocols of bias assessment.

The article presented in [2] looks into the empirical barriers to the deployment of AI in an ethical manner within companies, and lists the inability of accountability, non-transparency. It provokes the contrasts to the practices of development and ethical policies using a case study in the fields of

finance and AI in healthcare. It is also research that proposes mixed forms of governance however, it does not provide any substantial technical guidelines of implementation. The future research approach entails elimination of policy-practice gaps.

The research conducted in [3] suggests the usage of datasheets as the templates of standard documentation to improve the transparency of AI. The information about the content of datasets, data collection, proposed usage, and limitation will be needful in each datasheet. Test is validated based on research and industry data and enhances the sensitivity of bias. They have voluntary adoption and documentation maintenance issues. Research in the automated datasheet creation is in the future.

The authors of [4] recommend using model cards that are generic reporting frameworks of AI models and stipulate their functionality in demographics, intended application, and ethics. Experimented model cards which are proven on commercial ML models amplify the understanding of the stakeholders and provides fairness tests. The critical issues involve the issue of complex behaviour quantification of a model, and maintenance of a current documentation. Further studies on the issue are conducted on dynamic model cards and real-time measures.

In [5], the provided piece of work is the **NIST AI Risk Management Framework (AI RMF 1.0)**: which attempts to advance the conception of ethics throughout the project lifecycle, the design phase, as well as subsequent deployment. The structure has methodical processes of risk identification and evaluation and reduction of risks. It is very comprehensive though, its implementation needs a lot of commitment by the organization. The future work should be sector specific changes and lightweight models that target SMEs.

The paper outlines the IEEE 7000-2021 standard of managing the ethical issues involved in the course of the system design design in the [6], and gives a model process with stages and artifacts. The standard helps in developing a backtrack of the code of ethics to the technical requirements. The benefits of AI project validation in healthcare are increased ethical compliance and an increase in the documentation load. The next phase of work would be the integration and certification of the toolchains.

The paper by [7] creates detailed taxonomies of Explainable AI (XAI) techniques and classifies methods based on the type of explanation, the scope of explanation, and technical implementation. The taxonomy simplifies the choice of methods used depending on the application requirements and

needs of the stakeholders. It is theoretically sound but practically, a lot of expertise is needed. Future directions include the decision-support tool that would be used to select the XAI method.

In [8], the paper has suggested SHAP (SHapley Additive exPlanations) as a single model of explaining model predictions using cooperative game theory. The technique gives not only feature importance on a global basis but also on local instances. SHAP is more consistent than other approaches when evaluated on various ML models. Big data computation is still the challenge in large-scale deployment. Approximation algorithms are optimized in the future.

In [9], the study proposes LIME (Local Interpretable Model-agnostic Explanations) that can be used to explain individual predictions by approximating models that are complex, using interpretable surrogates that are locally defined. The method can be used with any access-denied model-internal classifier. Although they are flexible, explanations are unstable on the various random samples. Future work increases stability and gives estimates of confidence of the explanations.

In [10], the research supports the use of inherently interpretable models instead of post-hoc explanations in such critical tasks as healthcare and criminal justice. It contrasts interpretability-accuracy tradeoffs within classes of models, which discover interpretable models are frequently relevant to many high-stakes tasks. The strategy makes it easier to explain but could compromise performance in more complicated areas. Hybrid interpretable-deep architectures are developed in the future.

The research conducted in [11] officially identifies the concept of equality of opportunity as a measure of fairness where the true positive rates of the demographic groups are equal. The mathematical model allows measuring and optimization. Used on the dataset of hiring and lending, the metric will display the patterns of discrimination that would not be visible with the use of simpler metrics of fairness. Some of the implementation issues involve trusted collection of attributes. Future directions The proxy-resistant formulations are discussed in the future.

The article of [12] investigates the notion of demographic parity as a fairness criterion, that is, equal rates of selection between groups. It examines implementation strategies such as pre-processing method, in-processing method and post-processing method. Although conceptually simple, demographic parity may come into conflict with meritocratic principles in the competitive

environment. Context-conscious fairness standards are developed in the future that weighs various ethical aspects.

In [13], UCI Adult Income dataset is analyzed as a standard of fairness algorithms evaluation that images the features, drawbacks, and rightful applications. The data set allows fairness intervention to be compared across time on a standard basis but has historical biases of the inequities of society. These limitations have to be considered in conclusion made by researchers. The development of future work creates more representative benchmark datasets.

The [14] research uses the data of the COMPAS recidivism risk assessment and it reveals that the racial accuracy and false positive rates of prediction are very different. The review spawned a lot of discourse on the subject of algorithmic justice in law enforcement. Weaknesses of the datasets include the lack of representation of the features and ambiguous ground truth. Increased criminal justice risk assessment procedures are of interest to the future research.

The study within [15] documents the fact that even standard benchmark data like German Credit are biased in nature and hence result in discrimination unless corrected. It considers the sources of bias including the sampling procedures, labelling procedures, and historical iniquities. The study provides recommendations about the use of bias-aware datasets but mentions that complete debiasing might be impossible. Quantification of bias is formulated in future.

The researchers demonstrate in [16], that even after the technical fairness measures have been introduced in the AI systems, the historical biases were still present, which emphasizes the importance of socio-technical solutions that guarantee that the change in algorithms and organization is accompanied by the changes. Case studies is not much when it comes to the effects of technical fixes alone. The stance of the research is what requires a comprehensive fairness approach that is sensitive of the power structures and situations of decision making. The future work is developed on the basis of integrating socio-technical systems.

The study of [17] argues that explainability is a quantifiable non-functional requirement of processing of REs and proposes the measures of the different elements of the explanation like completeness, fidelity and comprehensibility. The systematic specification and validation of the



requirements of the explanations are made possible by the methodology. The practical implementation requires stakeholder education and tools support. The future of work is creating standardized measures of quality of explanations.

In [18], the research indicates that ethical requirements and relationships among ethical concepts in the AI systems should be modeled by using formal ontologies. The reasoning of the consistency and completeness of requirements is possible with the help of the ontology. Even though it is a theoretically sound approach, it takes much expertise to evolve ontology. The future work entails reusable ethical requirements domain ontology.

In [19], the study introduces E-LENS framework, which is founded on assurance case to form the relationship between the ethical requirements and the evidence and validation stage. The approach provides the formal argument of the ethical compliance but can become complex with the issue of large systems. The future work develops the support of tool assurance case management and automated evidence collection.

In [20], the paper proposes the use of capability-based approaches to realize high-level principles of Responsible AI as measurable system capabilities with the help of capability modeling and refinement. The strategy bridges the gap between abstract concepts and practical applications but it must be broken down into competencies. The patterns of capability formed in future work are based on common ethical demands.

The [21] study presents the problems practitioners face in seeking to capture, measure and validate ethical requirements based on the interview of 45 AI developers. These challenges are poor definition of ethical concepts, lack of measurements and resistance by corporate. The research outlines some localities of pain to which there are limited solutions. The future of practitioner-based tools and processes change.

A review of the literature of Requirements engineering of AI (Requirements engineering of artificial intelligence) is presented in [22] where it is demonstrated that lifecycle integration of ethical requirements has some significant gaps. According to the mapping, there is a disproportioned focus on technical considerations and no concern with ethics. The research provides the foundation on

certain research but does not address the gaps that are identified. The future study focuses on the unexplored areas of ethical RE.

As the study in [23] suggests, traceability is regarded as one of the important drawbacks of ethical RE strategies, and the majority of the approaches cannot have any plausible links between the ethical principles and the artifacts of implementation. Without traceability, people can never act ethically in a systematic manner. The paper raises the issue but offers very few solutions. In future, ethical requirement traceability structures are created.

In reference [24], the validation techniques of an ethical requirement are examined and the majority of the techniques do not have the capability of automated tools and are examined by hand. This constrains broadening and uniformity of moral validation. The methods of validation have been determined to be valid according to the research, but major integration issues are encountered. The integrated validation toolchains are developed in the future.

The article on [25] has given a roadmap on how to make responsible AI with the highlight on the governance systems, accountability mechanisms, and the coordination of the lifecycle. The roadmap also determines the transitioning route and the degree of maturity but does not give much information about how implementation can be executed. Future jobs strategize roadmaps in the industries and have tangible milestones.

The study [26] introduces the Software Engineering to Machine Learning (SE4ML) project that is intended to transform the traditional method towards the SE to the ML systems. It places exceptional obstacles on testing, debugging and maintains ML systems. The paper provides the best practices, which are not empirically justified. Longitudinal research of SE4ML is carried out in the future.

As outlined in the article [27], the writer is presenting the Fairlearn toolkit developed by Microsoft that quantifies and propagates the justice of AI among the demographics. The toolkit offers metrics, visualization and mitigation algorithms, which are scikit-learn compatible. Although it is simple, it has to be integrated into the current work processes, which sets the implication of adjusting to it. Future work extends the number of ways of mitigation and can be more scalable.

The article at [28] introduces the IBM AI Fairness 360 (AIF360) toolkit that has elaborate fairness measures and the algorithms included in the paper in its entirety in the ML pipeline. The toolkit facilitates the various definitions of fairness and also benchmarking. Adoption may not be attained due to complexity and computational requirements. APIs are made easier and performance optimization is realized in future.

In the article of [29], the focus is on the technical and organizational impediments in the implementation of such tools of ethics, as Fairlearn and AIF360 in the development processes. The complexity of equipment, incompetence and no process alignment are the major weaknesses. The trends of integration are set in the study, though not much information on adoption. Lightweight integration structures are developed in future.

The paper of [30] also supports the idea of the requirement level validation in contrast to post-hoc analysis and gives patterns of architecture to identify the requirement-tool integration. It is quick to get validated when it has to re-evaluate the development processes. Guidelines Future work designs refer to architecture and integration guidelines.

In [31], the study performed takes into consideration the user centric designs in the XAI where the user of the explanation is anchored to the needs and the cognition styles of various stakeholders. The user studies indicate that there exists significant discrepancy in the preference of various roles of explanations. The study has design principles, and no implementation structures. The adaptive explanation systems are developed in the future.

The current study [32] covers the impacts of various XAI strategies on human-AI interaction with regards to user trust and decisions. In the characteristics of explanation, they indicate when controlled experiments are used, the fact that they play an important role in proper calibration of trust. The study gives proof and limited suggestions in design. The future work involves the creation of patterns of explanation design that are evidence-based.

The writing in [33] suggests that documentation of the data sets (which must be in accordance with the datasheet standards) must be provided to the requirement management systems to accomplish the intent of ethical traceability. This allows linking the requirements and data properties and

constraints. It involves the lengthening of tools and modification of the processes. The future work has included documentation-management tools.

The research in [34] supports the idea of incorporation of model cards and the like documentation into the system of requirement management, the establishment of bidirectional traceability between the requirements and the model attributes. This aids the impact analysis in case of change of models. Implementation is technically challenging in terms of integration. The APIs of standard integration are developed in future work.

In [35], the paper suggests the use of runtime monitoring systems to evaluate the ethical adherence of deployed AI systems in the continuous collection and analysis of metrics. The method allows preventing the problem but demands the selection of metrics and setting of the threshold. Adaptive monitoring frameworks are developed in the future.

In [36], the study proposes the frameworks of post-deployment auditing of AI systems, such as periodic review, incident analysis, and impact assessment. To perform auditing successfully, one needs to have access to the system data and well-defined accountability frameworks. The study gives templates of audit but observes the variability in implementation. Future employment creates sector audit protocols.

The research of [37] indicates that the rate of adoption of ethical practices of RE in industry is low even with the increasing awareness and it shows that there are barriers both organizational, technical and cultural. A survey of 200 organizations indicates that there are policy/practice implementations gaps. The study measures adoption issues but does not provide many intervention measures. The future work makes development of adoption acceleration programs.

According to the study conducted in [38], the complexity and the absence of automation is one of the primary barriers to the implementation of ontology-based and assurance-case ethical frameworks. To be adopted practically, simplification and tool support is required. The study examines sources of complexity but offers little advice as to how things should be simplified. Future work creates simplified versions of the frameworks.

The research in [39] records that the industry has strong demand of practical, tool-based, approaches of ethical RE that integrate theory and practice. The surveys conducted in the industry focus on usability, integration, and visible ROI as opposed to theoretical completeness. The research identifies requirements for practical tools but doesn't develop solutions. Future work focuses on industry-aligned tool development.

The study in [40] analyzes EU GDPR Articles 13-15 requirements for transparency and the "right to explanation" in automated decision-making systems. The research interprets legal requirements in technical terms and identifies implementation approaches. Legal interpretations continue to evolve. Future work monitors regulatory developments and develops compliance frameworks.

### **Datasets Description**

#### **Dataset 1: Adult Income Dataset from UCI**

This data consists of the demographic data of 1994 U.S census age, gender, income, occupation, and education. The study will aim to examine the prejudice and integrity of income forecasting models. The racial or gender prejudice is often researched with the help of the dataset, which is why it is an attractive fact that supports the validation of ethical requirements.

#### **Dataset 2: COMPAS Recidivism Dataset**

The COMPAS algorithm process is based on the behavioral, demographic, and criminal history of the dataset to predict a potential risk of a person recidivating. The objective of the study is to demonstrate how ethical conceptual principles such as fairness and transparency can be implemented in AI-based decision-making systems by experimenting with bias and accountability indicators within the framework of the RE.

#### **Dataset 3: German Credit Dataset**

The information about the job status, credit history, and financial situation of people is used in this dataset that is utilized to estimate credit risk. It will help in evaluating the requirements of explainability, privacy and fairness in the financial decision making systems. It is possible to substantiate the moral standards related to responsible lending and equal treatment due to the dataset.

## Dataset 4: Bank Marketing Dataset (UCI Repository)

This dataset gives the demographics of the clients and the outcome of the marketing campaign of a Portuguese banking institution. The purpose of the study is to show how a clear decision logic could make the consumer more trusting of AI-based marketing by testing explainability and accountability criteria..

**Table 1: Datasets Description**

| Dataset Name                                    | Size               | Features                                                                                                       | Protected Attributes                                   | Target Label                    | Purpose in Research                                                                                                                                                                                              |
|-------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adult Income Dataset (UCI Census Income)</b> | <b>48,842 rows</b> | age, workclass, education, marital-status, occupation, relationship, race, sex, hours-per-week, native-country | <b>Race, Gender</b>                                    | <b>income</b><br>(>50K or ≤50K) | <ul style="list-style-type: none"> <li>• Test gender &amp; racial bias in income prediction using fairness metrics.</li> <li>• Validate fairness, transparency requirements using SHAP, LIME, AIF360.</li> </ul> |
| <b>COMPAS Recidivism Dataset</b>                | <b>7,000+ rows</b> | age, age_cat, priors_count, juv_fel_count, juv_misd_count, charge_degree, days_b_screening_arrest              | <b>Race, Gender</b>                                    | risk score<br>(low/medium/high) | <ul style="list-style-type: none"> <li>• Known biased dataset → ideal to demonstrate fairness violations.</li> <li>• Used to evaluate fairness metrics and mitigation in your RE framework.</li> </ul>           |
| <b>German Credit Dataset (UCI)</b>              | <b>1,000 rows</b>  | credit history, purpose, employment status, personal-status, housing, savings, age                             | <b>Age, Gender</b><br>(encoded inside personal-status) | credit risk<br>(good/bad)       | <ul style="list-style-type: none"> <li>• Test discrimination in credit scoring.</li> <li>• Validate fairness and accountability requirements.</li> <li>• Good for SHAP explainability demonstrations.</li> </ul> |

|                                     |                    |                                                                          |                |                                               |                                                                                                                                                                                                         |
|-------------------------------------|--------------------|--------------------------------------------------------------------------|----------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bank Marketing Dataset (UCI)</b> | <b>45,211 rows</b> | age, job, marital, education, balance, housing, loan, campaign, duration | Age (Only one) | subscribe<br>d to term<br>deposit<br>(yes/no) | <ul style="list-style-type: none"> <li>• Used to test fairness when dataset has limited protected attributes.</li> <li>• Good for demonstrating transparency and accountability (SHAP/LIME).</li> </ul> |
|-------------------------------------|--------------------|--------------------------------------------------------------------------|----------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Table 2: Tentative Timetable**

| Process        | Sep, Oct                   | Nov, Dec          | Jan         | Feb, Mar | Apr, May              | June          |
|----------------|----------------------------|-------------------|-------------|----------|-----------------------|---------------|
| Planning       | Topic selection<br>+ Intro |                   |             |          |                       |               |
| Documentation  |                            | Literature Review |             |          |                       |               |
| Implementation |                            |                   | Development |          |                       |               |
| Development    |                            |                   |             | Coding   |                       |               |
| Debugging      |                            |                   |             |          | Debugging<br>+ Thesis |               |
| Testing        |                            |                   |             |          |                       | Thesis Review |

### **Potential Impact and Contribution to Scientific Community and Society**

This study will be valuable to the scientific community because it provides a systematic structure that considers ethical implications in software requirements engineering of AI systems. It will present a systematic approach towards identifying, documenting and attesting moral norms such as accountability, openness and equity. Also, the research will contribute to the development of an online validation tool that will enable developers and researchers to check the level of ethical compliance when developing systems.

The study will focus on enhancing the creation of more responsible and reputable AI systems by the society. The given solution will contribute to the minimization of such concerns as prejudice, discrimination, and the absence of transparency in the AI decision-making process by considering moral aspects in its early development. Lastly, the work will help create the AI systems, which will not harm human values, cause people to trust them more and develop technologies in a sustainable and ethical manner..

### **One Most Relevant Industry**

Steller Stack

### **One Targeted Product (Name, Type, Description)**

**Product Name:** EthicalReq-AI

**Type:** Requirements Ethical Requirement Engineer and Requirement validation tool provided as Web-based requirements.

EthicalReq-AI web-based platform is supposed to support the identification, documentation, and certification of the ethical needs in artificial intelligence (AI) systems. Researchers and developers in the platform can be in-capable of realizing ethical concepts such as accountability, explainability, openness, and fairness and align them with relevant models, datasets and metrics of assessment. It will also offer automatic validation tests all through the AI development lifecycle in order to assure that ethical standards will be attained at all times.

In addition, the site will have compliance reporting tools, ethical risk assessments tools and requirement traceability tools. All these features will empower the EthicalReq-AI to help enterprises create credible and ethical solutions with references to AI by bridging the gap between the theoretical ethical models and the use of AI.

### **Current Status of the Product**



In Progress

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## **Certificate of Originality**

I \_\_Sabahat Siddique\_\_, Reg. No. \_\_M24F0023AI003\_\_, MS Scholar at School of Computing Sciences, certify that my MS research proposal with title “\_\_Requirement

Engineering for AI Systems\_\_” declare the originality of contents that has been verified through plagiarism software and is \_\_\_\_%.

**Student Name:** Sabahat Siddique

**Reg. No:** M24F0023AI003

**Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Supervisor**

**Name:** \_\_\_\_\_

**Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_

### **Recommendation by the Research Supervisor**

I have read the synopsis entitled

\_\_\_\_\_

and agree to supervise Mr./Ms. \_\_\_\_\_ for the partial requirements of the degree

of MS in \_\_\_\_\_. The proposed project is academically, logistically,

administratively, and financially feasible and all the required literature, equipment, laboratories,

space and transport facilities are or will be available at PAF-IASST, Haripur or at

\_\_\_\_\_.

The following facilities/finances will be required for this project:

Financial Requirements:

Estimated Cost: Rs. \_\_\_\_\_

These financial requirements will be fulfilled from \_\_\_\_\_ source(s).

Supervisor's Name: \_\_\_\_\_ Signature \_\_\_\_\_ Date \_\_\_\_\_

### **Signed by GEC Member**

| S. # | Name of Committee Member | Designation | Signature and Date |
|------|--------------------------|-------------|--------------------|
| 1    |                          |             |                    |
| 2    |                          |             |                    |
| 3    |                          |             |                    |
| 4    |                          |             |                    |

### **Recommendation by Departmental Graduate Committee (DGC)**

Certified that the synopsis has been seen by members of DGC and considered it suitable for putting up to the Advanced Studies and Research Board of PAF-IASST (ASRB).

Graduate Coordinator: \_\_\_\_\_ Signature: \_\_\_\_\_ Date: \_\_\_\_\_

Convener DGC (Chairman/HoD): \_\_\_\_\_ Signature: \_\_\_\_\_ Date: \_\_\_\_\_

### **Dean, Faculty of .....**

\_\_\_\_\_ Recommended for placement before ASRB.

\_\_\_\_\_ Not Approved on the basis of following reasons

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

**Secretary ASRB**

\_\_\_\_ Approved from ASRB.

\_\_\_\_ Not Approved on the basis of the following reasons

Signature \_\_\_\_\_

Date \_\_\_\_\_